Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **Uh?**
b. Team leader name: **Jaewoo Choi**
c. Problem Statement: Live-commerce hosts speak unscripted for hours. A wrong price or a claim the regulations forbid ("this cream treats eczema") can reach viewers before anyone corrects it.

Brief about the idea

It checks what the host says, while they say it, and clears when they fix it

A show on the deployed app (10/11), lines typed into the live judging path. Each final line goes to four Gemini judges at once.

"And honestly, this cream treats eczema, so your eczema will be gone."	**Compliance judge** Disease claim: ASEAN cosmetic claims guideline, App. III 2.d	**Red box in 1.3 s** "Sorry, I take that back. Helps relieve the feeling of dryness."
"Today's live price is just S$32."	**Fact checker** The prompter shows S$36.00	**Orange box in 0.8 s** The live price is S$36.00.
"Sorry, I take that back. It helps relieve the feeling of dryness." · "My mistake, the live price is S$36."	**Same judges, next lines** The claim is withdrawn; the right price is said	**Both boxes clear** in the host's own words
"It keeps your skin hydrated for up to 24 hours, in our in-house test." · "Okay, next, let me show you the ampoule."	**Prompter and context judges** Full point said · "next" + the next product's category	**Point ticked · prompter moves on** no click by the host

Viewer questions: answered automatically when the product record answers them ("how much is it today?" got the live price in 4.3 s); health questions ("I get redness easily...") go to the host's screen with a suggested reply.

Opportunities

Checks against the regulation text, and the host fixes it on air

Why now

Market

25%
video commerce share of SEA e-commerce (2025, forecast)[1]

~600M
live-commerce users in China (2024)[2]

Risky claims on air

30 / 120
Korean live shows with suspected unfair ad claims (2020)[3]

Rules now ask for it

2026
China: misstatements said on air must be corrected on the spot (Art. 31)[4]

How different is it from any of the other existing ideas?

Common approach	What it leaves out	Ours
Keyword lists of banned words	Words, not meaning: negation, rephrasing, quotes	Gemini reads the whole line against the regulation PDF itself
Compliance review after the show	The claim has already gone out	A red box in about 1-4 s, with a line to say instead
Approving the script before air	Hosts improvise; prices change	Checks unscripted speech and today's prices, live

How will it be able to solve the problem?

For brands and platforms that run their own live shows. The host reads a red or orange box from a distance, says the line shown, and the box clears. A solo host needs no operator; an operator can watch the same show.

USP of the proposed solution

Models judge meaning; code never overrules them. Any product enters through its own documents, with no product-specific prompts, and the correction is checked in the host's own words.

[1] Google, Temasek, Bain, e-Conomy SEA 2025 (forecast): blog.google/.../sea-economy-2025 [2] CFIS & CIECC, China New E-commerce Development Report 2025: cac.gov.cn/2025-07/28
[3] Korea Consumer Agency, Live Commerce Advertising Survey (2020): kca.go.kr (PDF) [4] SAMR & CAC, Livestream E-commerce Supervision Measures, Art. 31 (2026): cac.gov.cn/2026-01/07

List of features offered by the solution

Before air people approve, on air the host only speaks, after air a recap

Before air

- Upload a product PDF; no fixed template to fill in
- Gemini reads it into a product record: brand, category, prices, prompter points, speech terms
- Upload the regulation PDFs
- A person approves; nothing else is typed

On air: what the host says

- Red box: a claim a regulation clause forbids, the clause, a line to say instead
- Orange box: a price or fact that differs from the prompter
- Boxes clear when the host corrects it in any words
- Prompter ticks a point only when its full meaning is said; "next" + name or category moves to the next product

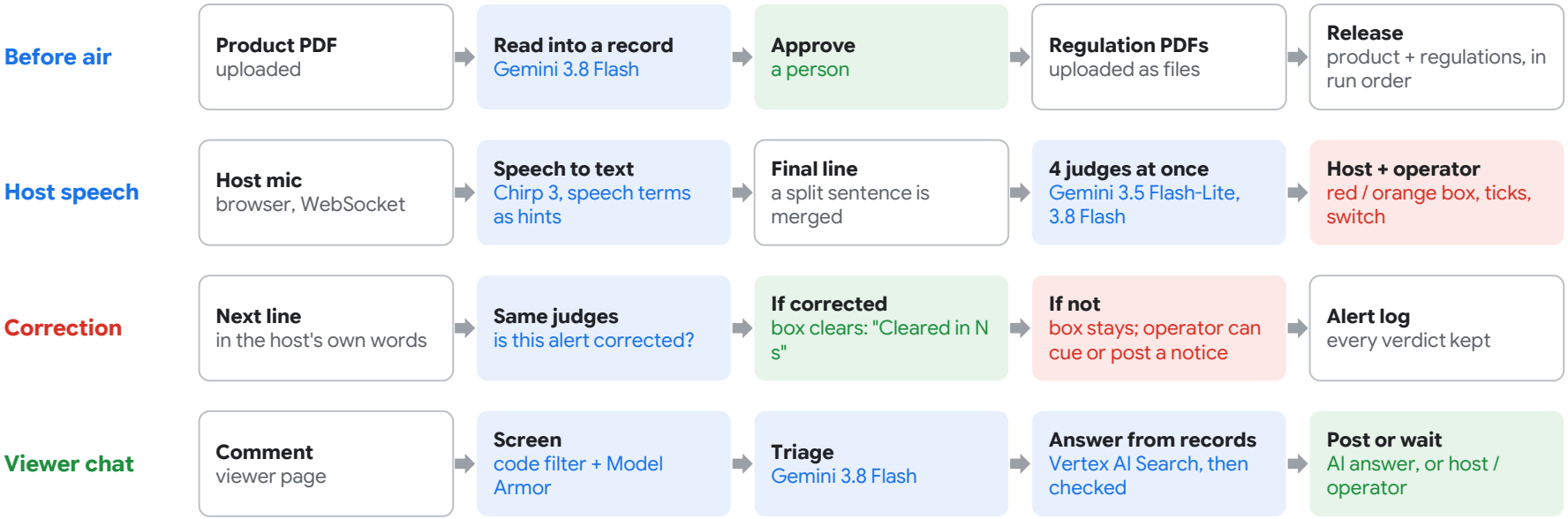
On air: what viewers ask

- Spam, abuse and prompt attacks screened (Model Armor)
- Answered automatically when the product record answers it, marked "AI answer"
- Health, claims and test results wait for the host or the operator, with a suggested reply

After air: an AI recap from the show's numbers (alerts and how long each took to fix, unanswered questions, missed points), exported as PDF or JSON.

Process flow diagram or Use-case diagram

One approval before air sets the evidence every judge uses

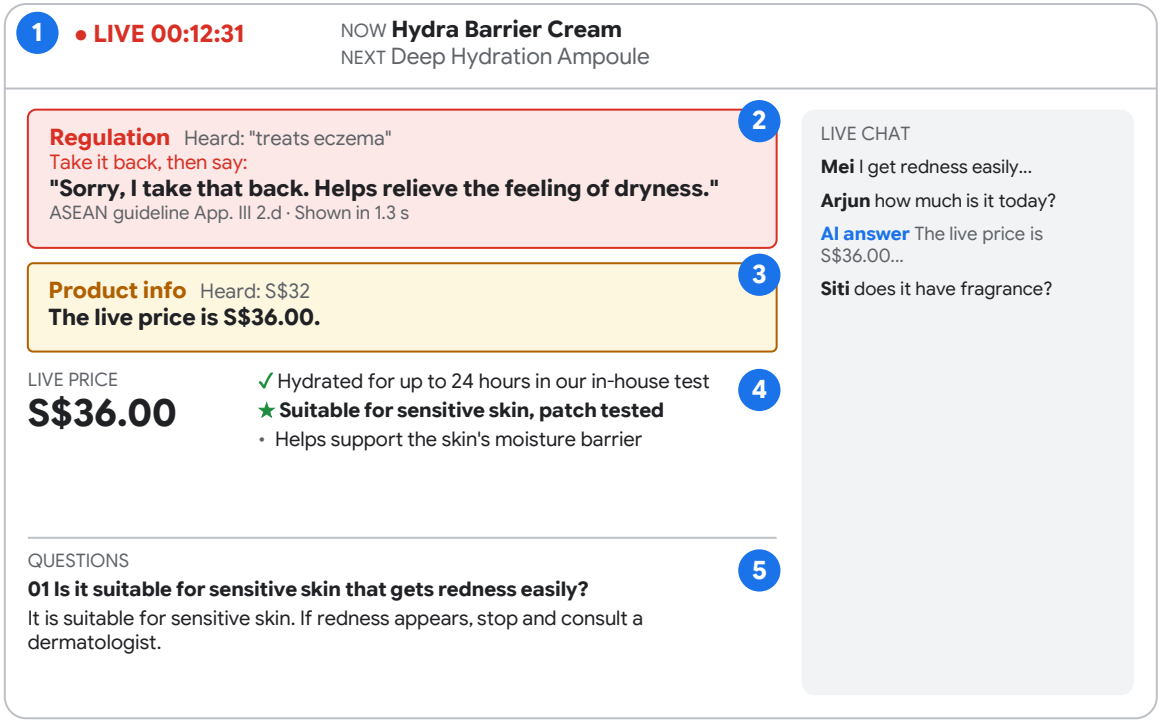


Legend blue = Gemini green = a person or the result they see red = what the host must fix

The judges read only what was approved: the regulation files for red, what the prompter shows for orange. Nothing is typed in by hand during the show.

Wireframes/Mock diagrams of the proposed solution (optional)

Alerts, the line to say and the talking points on one screen



1. Show clock and the product on air, with the next one
2. Red: the regulation clause and one full sentence to read
3. Orange: the right value from the prompter
4. Prompter: ticks fill in as points are said
5. Viewer questions that need a person, with a reply

Layout schematic. The host never clicks; real screens on slide 10.

Architecture diagram of the proposed solution

One Cloud Run service runs each show and calls managed Google Cloud AI

Inputs and screens

Host mic
audio over WebSocket

Viewer chat
comments

Product + regulation PDFs
uploaded before air

Screens
host, operator, viewer, summary (SSE)

Cloud Run service (FastAPI)

STT stream per show · a split sentence is merged · timers

Compliance
Flash-Lite · regulation PDFs as files

Fact checker
Flash-Lite · what the prompter shows

Context
Flash-Lite · switch, question hold

Prompter
3.8 Flash · full-meaning ticks

4 judges in parallel on every final line

- Comment path: screen, triage, answer, check
- Code maps ids, runs timers, draws screens; it never drops a verdict
- IAP login · up to 3 instances · session affinity

Google Cloud managed services

Speech-to-Text V2 chirp_3, region us

Vertex AI Gemini 3.5 Flash-Lite · 3.8 Flash (global)

Vertex AI Search product sheets for viewer answers

Model Armor screens viewer comments

Firestore show state, alerts, every AI call

Cloud Storage products, regulations, releases (write-once)

Each judge has one source of truth: regulation files for red, the prompter for orange. Show state and every AI call are stored in Firestore.

Technologies to be used in the solution

Each Google Cloud service has one job in the show

Service	Model or setting	Its job
Speech-to-Text V2	chirp_3, region us, STANDARD endpointing	Streams the host's speech; speech terms from the product record as hints
Vertex AI Gemini	3.5 Flash-Lite	Compliance judge, fact checker, context judge (one call each per line)
Vertex AI Gemini	3.8 Flash	Prompter judge; reading product PDFs; comment triage, answers, recap
Vertex AI Search	product sheets, per market	Finds the product-sheet passage behind a viewer answer
Model Armor	before any model	Screens viewer comments for spam, abuse and prompt attacks
Cloud Run + IAP	us-central1, up to 3 instances	FastAPI backend, React screens, server-sent events
Firestore	Native mode	Sessions, transcripts, alerts, segments and every AI call with its tokens
Cloud Storage	write-once objects	Approved products, regulation files and releases
App stack	Python 3.12, FastAPI, React 18, TypeScript, Vite	Agent Development Kit for the AI PD agent; pytest, Vitest, Playwright

Estimated implementation cost (optional)

Judges + speech: about US$3.7 an hour at 600 lines an hour

Item	Basis (measured)	Per line	Per hour (600 lines)
Compliance judge	Flash-Lite, ~16.8k tokens (regulation PDFs), cached	$0.0011	$0.63
Context judge	Flash-Lite, ~4.0k tokens in	$0.0016	$0.94
Prompter judge	3.8 Flash, ~1.2k tokens in, low thinking	$0.0012	$0.74
Fact checker	Flash-Lite, ~5.3k tokens, part cached	$0.0008	$0.45
Speech-to-Text	chirp_3, $0.016 per audio minute		$0.96
Total	judges per line; per hour adds speech	**$0.0046**	**≈ $3.7**

How it was measured

Token counts of every judge call in 2 shows on the deployed app (10/11, 14 host lines) from Firestore, times Google's list prices. Per hour assumes 10 final lines a minute (600 lines, an assumption, not measured).

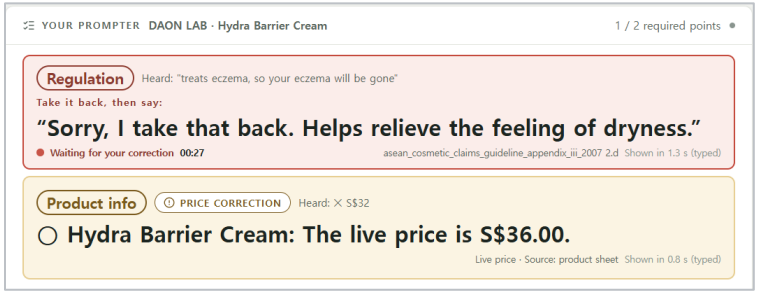
Not included

Repeat requests sent when a judge is slow (not logged; at most doubles that call). Viewer comments: ~$0.006 a comment for triage. The recap: $0.007 a show. Cloud Run and Firestore. 3.8 Flash list price doubles on 2027-01-01.

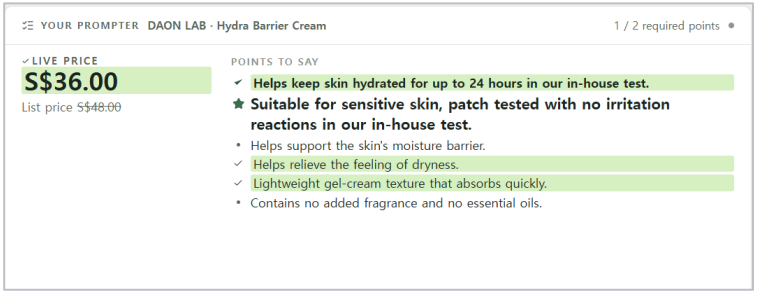
List prices on 10/11: Gemini API pricing page (ai.google.dev) and Speech-to-Text pricing (cloud.google.com).

Snapshots of the prototype

One show on the deployed app, from alert to correction



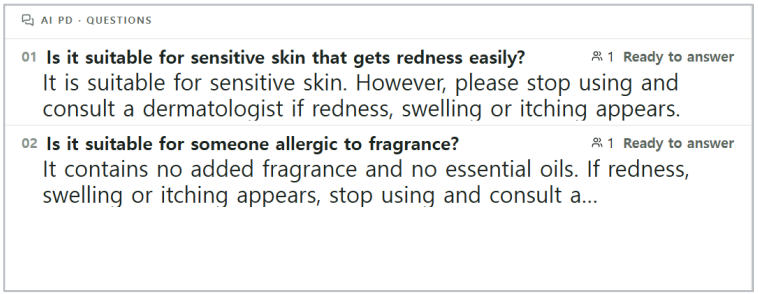
1 Red and orange boxes on the host screen



2 Both cleared in the host's words; points ticked



3 Operator: the clause, the AI summary, actions



4 Viewer questions that need the host, with replies

Cloud Run revision 00028, 10/11. Lines typed into the live judging path (marked "typed"); speech goes through Chirp 3 to the same judges (measured on slide 11).

Prototype Performance report/Benchmarking

Detection 94.0% on a fresh set; false alerts still above the 1% line

Detection
94.0%
retest after one change: 95.5%
Pass line ≥ 95%

False alerts on safe lines
2.4%
retest: 1.6% (2 of 127)
Pass line < 1% · not met

Wrong strongest alerts
0
too-strong alert · both runs
Pass line 0

Alert latency p95
4.2 s
end of line to host screen
Pass line ≤ 5 s

Sealed set 7: 6 products the app never saw, invented documents; 261 host lines spoken by a TTS voice through the real Chirp 3 path (27 of 288 left out: two hosts at once, stress lines, rules outside the pack). Labels by GPT and a blind Claude critic. Per product 82.6-100%.

Why models judge, not code

Same STT lines (set 5)	False alerts	Wrong strongest
Code rules on top of the model	**19.4%**	5
Model verdicts only	**0.9-1.9%**	1

Detection on each new unseen set

84% → 91.0% → 89.6-90.3% → **94.0%**

sets 3, 4, 6, 7 (set 6 was the first through TTS and STT)

Other measured parts

Test	Result
Prompter judge, partial vs full points (10/11 probe)	wrong ticks 0 / 40, missed 0 / 63
Product switch on a category word only	0 / 5 → 4 / 4 with category
Viewer comments: question / reaction / hide (233)	99.6%
Comments: auto vs needs a person, as labelled	90.6% · injection replies 0

The submitted revision has not yet been run on a fresh sealed set; no real human host yet.

Additional Details/Future Development (if any)

Flags people act on; next, a fresh test of the submitted version

Responsible AI (in the prototype)

- A red box marks a claim for the host to fix. It is not a legal ruling.
- The line to say instead is built only from the product's approved claims.
- Health, claim and test-result questions wait for a person; automatic replies say "AI answer".
- Judge calls are stored with input, output and tokens; code never drops a verdict.
- No product-specific prompts: a new product enters only through its documents.

Next (planned, not in the prototype)

- Freeze the judges, then run a new sealed set on the exact submitted revision.
- Test with a real human host and two hosts speaking over each other.
- Bring false alerts under 1% for every product.
- Log repeat requests so the cost per show is complete.
- Copy AI-call logs to BigQuery for analysis across shows.

Provide links to your:

Try it: start a show, say a claim, then take it back

GitHub Public Repository

github.com/jwchoiKR/ai_builder_cup_uh

Demo Video Link (3 Minutes)

to be added

Final Product Link

to be added

Three steps for an evaluator

1. **Open the hub and start a show from an approved product**
2. **Say "This cream treats eczema" and watch the red box**
3. **Read the line it shows and watch the box clear**

Every alert comes from the live models on the same path as a real show; nothing is pre-stored. Speak into the browser mic on the host screen.

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

